

Lorenzo Agabiti

PhD candidate in Mathematics at Sorbonne University, with international experience.

[LinkedIn](#) [GitHub](#) [Contact](#)

EDUCATION

- 10/2024 – present** **PhD in Stochastic Differential Equations, Sorbonne University**
Paris, France
 - Winner of the MathPhdInFrance scholarship (Marie Skłodowska-Curie actions cofund).
 - Co-supervised between Sorbonne University and Nancy University, by Lorenzo Zambotti (LPSM) and Antoine Lejay (IECL).
 - Project: *Existence, uniqueness and a priori estimates for solutions à la Davie of rough equations with arbitrary Hölder regularity of the driving path.*
- 09/2021 – 02/2024** **MSc in Applied Mathematics, EPFL**
Lausanne, Switzerland
 - Minor in Financial Engineering.
 - Coursework: Introduction to Stochastic PDEs, Theory of Stochastic Calculus, Stochastic Processes, Machine Learning for Finance, Numerical Integration of SDEs, Advanced Derivatives, Martingales in Financial Mathematics.
 - Project in Stochastic Real World Pricing: continuous financial markets, growth optimal portfolio, Bessel processes, minimized market model.
- 09/2023 – 02/2024** **Master thesis in Stochastic PDEs, Sorbonne University**
Paris, France
 - Carried out in the probability, statistics and modelling laboratory (LPSM).
 - Project on the theory of rough paths and the analytical theory of controlled equations with a very rough driving path.
 - Study of the theory of regularity structures and of the resolution of stochastic PDEs.
- 09/2018 – 07/2021** **BSc in Mathematics for Engineering, Politecnico di Torino**
Turin, Italy
 - Coursework: Real Analysis I and II, Probability Theory, Programming and Scientific Calculus, Applied Statistics, Linear Algebra, Numerical Analysis, Functional Analysis.
 - Final project: Krein–Milman’s theorem, a topic in functional analysis with applications to convex optimization.

WORKING EXPERIENCE

06/2023 –
09/2023

Quantitative summer internship, Credit Suisse

Zurich, Switzerland

- Credit Portfolio Modelling team: worked with the Credit Erc model and the internal statistical model for credit risk in Lombard and non-Lombard deals.
- Developed an application that approximates the impact of a new deal on the Value at Risk and Expected Shortfall of the bank portfolios.
- Skills used: statistics, R, Shiny, front-end programming (HTML, CSS, JavaScript).

RESEARCH

2026

Remainders of generalised Taylor expansions and a priori bounds for rough differential equations

With Alberto Bonicelli and Lorenzo Zambotti. arXiv preprint.

LANGUAGES

Italian	Mother tongue
English	Fluent
French	Intermediate
Spanish	Intermediate

IT SKILLS

Matlab	Advanced
LaTeX	Advanced
Python	Intermediate
R	Intermediate
C/C++	Intermediate
GitHub	Basic
HTML	Basic

EXTRACURRICULAR

2018 – 2019

Production manager, “Junior Achievement” start-up project

Rieti, Italy

- Managed the production and development of a new board game, which won the prize for best-selling product.

2016

“IMUN”, simulation of the United Nations working sessions

Rome, Italy

- Worked as a diplomat with delegates from around the world, practising negotiation and public speaking in English.